WhatsApp-Integrated Smart Taxi Booking and Real-Time Ride Management System

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINNERING

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DECLARATION

The Mini Project Report entitled "**K. Venkateswari** "is a record of Bonafide work of G. Manikanta-2420030366, I. Vishnu Vardhan – 2420030513, G. Tanay Tej - 2420030560 submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is certify that the mini project-based report entitled "**Project Title"** is a bonafide work done and submitted by G. Manikanta-2420030366, I. Vishnu Vardhan – 2420030513, G. Tanay Tej - 2420030560 in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Course Co-ordinator **Signature of HOD**

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ABSTRACT

The WhatsApp-Integrated Smart Taxi Booking and Real-Time Ride Management System is a smart transportation solution designed to make taxi booking easier, faster, and more convenient for users. The system integrates WhatsApp with a taxi booking platform, allowing users to book rides without installing or using a separate mobile application.

Users can provide their pickup location, destination, and other booking details through WhatsApp. The system processes the request, identifies available drivers, and assigns a suitable driver based on availability and location.

The system also provides real-time ride management, allowing users to receive booking confirmation, driver details, vehicle information, ride status, and notifications through WhatsApp. Drivers can manage assigned rides and update their status, such as accepting a ride, reaching the pickup location, starting the journey, and completing the trip.

An admin dashboard can be used to manage users, drivers, bookings, and ride information. The system can also support location tracking, fare calculation, and customer feedback.

Overall, the proposed system aims to provide a simple, efficient, and user-friendly taxi booking experience by combining WhatsApp communication, automated booking, driver management, and real-time ride tracking in a single platform.

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INTRODUCTION

The **WhatsApp-Integrated Smart Taxi Booking and Real-Time Ride Management System** is designed to provide a convenient taxi booking experience through WhatsApp.

Instead of requiring users to download and use a dedicated taxi application, the system allows them to interact with the taxi booking service through **WhatsApp**. Users can initiate a conversation, provide their pickup location and destination, and receive booking-related information directly through the messaging platform.

The system processes the booking request and identifies suitable drivers based on their availability and location. Once a driver is assigned, the user can receive driver details, vehicle information, booking confirmation, and ride status updates.

The system also supports driver-side ride management. Drivers can accept assigned rides and update the ride status during different stages of the journey, including:

- Ride accepted
- Driver heading to pick up location
- Driver reached pickup location
- Ride started
- Ride completed

An administrative interface can be used to manage users, drivers, bookings, and ride information.

The proposed system combines **WhatsApp communication, automated taxi booking, driver assignment, ride management, and real-time status updates** to create a unified transportation platform.

LITERATURE SURVEY

1. Uber – WhatsApp Ride Booking

Uber allows users to book rides through WhatsApp without opening its mobile application. Users can provide trip details and receive booking-related information. This simplifies the booking process and improves accessibility. It serves as a reference for integrating taxi booking with WhatsApp.

2. MakeMyTrip – WhatsApp Cab Booking

MakeMyTrip uses WhatsApp to simplify cab reservations and customer communication. Users can share travel details and receive booking assistance. This provides ideas for conversational booking and customer support.

3. Greentrip – EV Cab Booking

Greentrip uses WhatsApp integration to automate electric cab bookings and driver coordination. It demonstrates how automated messages and backend services can simplify ride management.

4. Sakhi Cabs – WhatsApp Integration

Sakhi Cabs combines a cab-booking platform with a WhatsApp booking assistant. It provides a reference for connecting customer requests with booking records and driver management.

5. Infobip – Uber Chatbot

Infobip developed a chatbot that enables Uber customers in Delhi-NCR to book rides through WhatsApp. It demonstrates conversational automation and multilingual customer interaction.

6. WhatsApp Business Platform

Meta provides APIs for sending and receiving messages and processing webhooks. These capabilities support automated booking confirmations and real-time ride-status notifications.

7. Twilio WhatsApp API

Twilio enables applications to exchange WhatsApp messages through APIs and webhooks. It can support chatbot development, booking requests, and automated customer notifications.

8. Uber Messaging Integration

Uber uses WhatsApp for customer communication and service-related notifications. This provides a reference for improving communication between customers and transportation services.

Problem Statement and Objectives

Problem Statement: Traditional taxi-booking systems often require separate applications, making booking and real-time ride management less convenient.

Problem Description

Customers face difficulties in booking taxis and tracking ride status efficiently. This project integrates WhatsApp with a taxi-booking system to simplify bookings, driver coordination, and ride updates.

Objectives:

- **Provide taxi booking through WhatsApp.**
- **Eliminate the need for users to install a separate taxi booking application.**
- **Allow users to provide pickup and destination details easily.**
- **Automatically process and manage taxi booking requests.**
- **Identify and assign suitable available drivers.**
- **Provide real-time ride status updates.**
- **Allow drivers to manage assigned rides.**
- **Provide driver and vehicle information to users.**
- **Provide an admin interface for managing the overall system.**
- **Support location tracking and fare calculation.**
- **Improve convenience and efficiency in taxi booking and ride management.**

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- Laptop/Desktop with a minimum of 4 GB RAM.
- Dual-core processor or higher.
- Stable internet connection.

Software Requirements:

- Frontend: React.js, HTML, CSS, JavaScript.
- Backend: Python and a database such as MySQL.
- WhatsApp Business API for messaging and booking integration.
- API's used: WhatsApp cloud meta API, real time data from Uber services,etc.
- Visual Studio Code for development.
- Google Chrome or any modern web browser for testing.

IMPLEMENTATION

Implementation Overview

The system is presented as a web-based taxi and transit booking prototype that connects a customer-facing booking interface with ride-management functions. The interface shown in the Results section demonstrates the customer conversation screen and a Hyderabad transit booking form.

Frontend Implementation

The frontend provides navigation for WhatsApp Customer, Driver App, Live Booking, and Admin Map. The customer interface displays a welcome message and language choices. The booking form allows users to enter a pickup station and destination, select the number of passengers, choose from suggested locations, and continue to search for available taxis.

User Interaction

The interface is designed to collect trip information in a clear sequence. After the pickup point, destination, and passenger count are selected, the user can proceed to the next booking step. The separate driver and admin sections provide entry points for ride handling and overall booking supervision.

Backend Implementation

The backend workflow is responsible for processing booking details received from the interface, checking that the required fields are provided, and maintaining the booking status. It can coordinate the availability of taxis and drivers, associate a suitable driver with a booking, and make the booking information available to the customer, driver, and administrator.

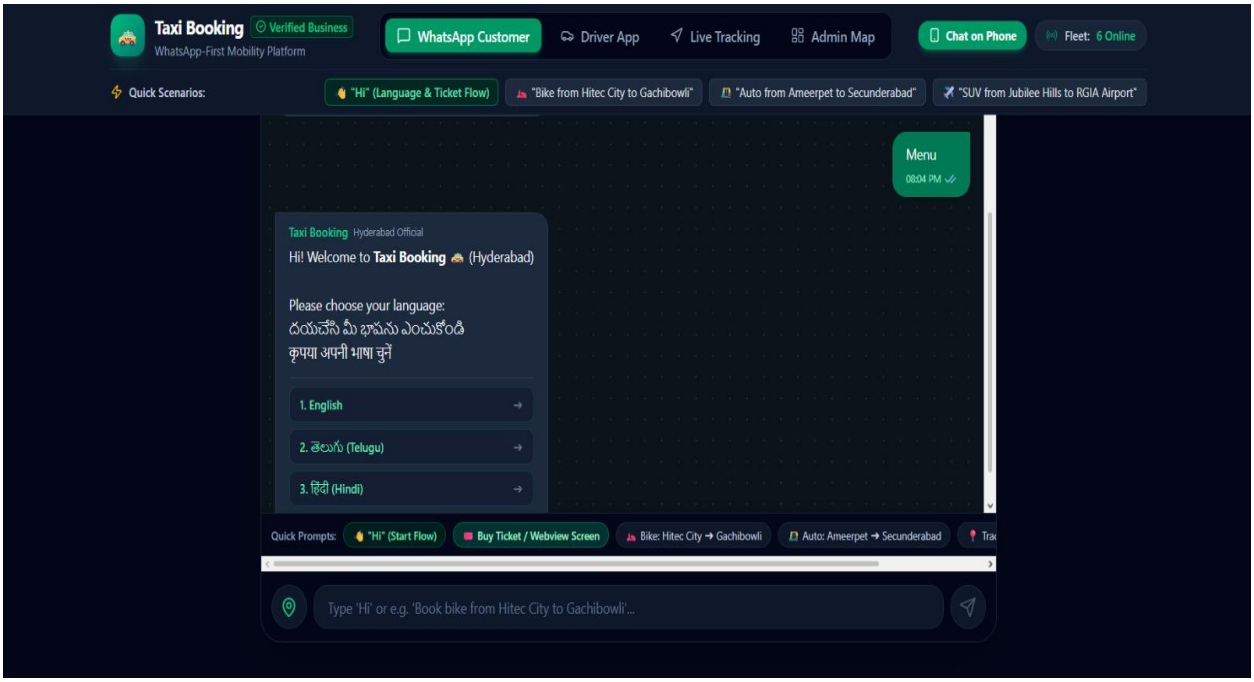
Ride Management Workflow

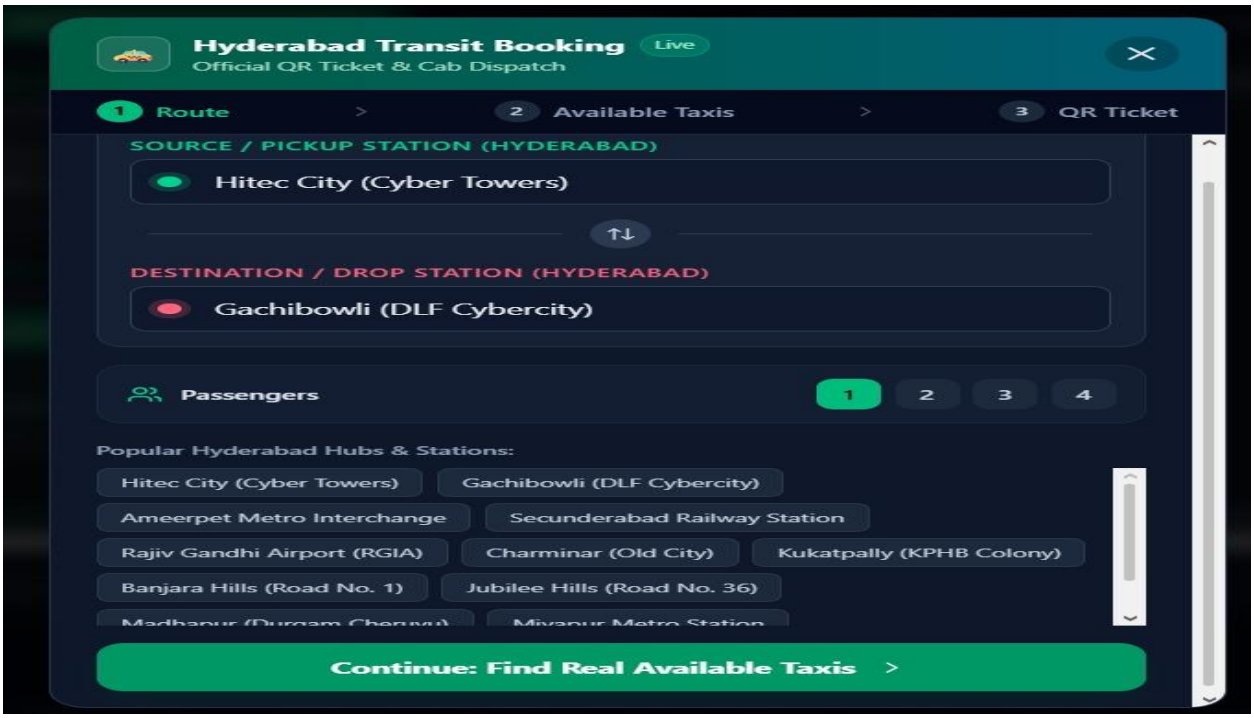
A typical workflow begins when a customer enters the pickup and drop-off locations and passenger count. The request is validated and checked against available taxis. After a suitable taxi is identified, the booking record and ride status can be updated as the trip progresses, with confirmations and status messages returned to the customer through the connected messaging channel.

Implementation Note

The screenshots in the Results section demonstrate the customer chat and transit booking interface. The report does not specify the exact programming languages, database, hosting environment, or live API configuration; these should be added only if they match the technologies used in the actual project.

RESULTS & DISCUSSION





The user can book the cab using WhatsApp by chatting with a chatbot.

CONCLUSION & FUTURE SCOPE

The WhatsApp-Integrated Smart Taxi Booking and Real-Time Ride Management System aims to simplify taxi booking through WhatsApp. It helps automate booking requests, driver coordination, and ride-status updates, providing users with a convenient and efficient booking experience.

FUTURE SCOPE

- Short-term: Improve booking confirmations and automated notifications.
- Medium-term: Integrate live GPS tracking, online payments, and driver ratings.
- Long-term: Implement AI-based route optimization, dynamic fare calculation, and predictive demand analysis**.**

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